



Experimental and numerical studies on the two “lock-in” regions characteristic of vertical vortex-induced vibration of Π -shaped composite bridge deck

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ABSTRACT

The Π -shaped bridge deck may easily occur vertical vortex-induced vibration, due to the bluff body aerodynamic configuration, which may cause the uncomfortable of vehicles and pedestrians on the bridges. The present study mainly focused on the two “lock-in” regions characteristics of vertical vortex-induced vibration (VIV) of Π -shaped bridge deck, which can reveal the aerodynamic mechanism of vibration mitigation countermeasures. The vertical VIV response of original Π -shaped deck and those with different countermeasures are experimentally investigated detailly. Computational fluid dynamics (CFD) method is employed to numerically simulate the surrounding flow field of original girder section and those with different aerodynamic countermeasures. The numerical method is firstly verified through the comparison with the results of corresponding experimental results obtained in wind tunnel test. Afterwards, the flow pattern and vortex structure around the girder section simulated by CFD are utilized to interpret the effects of the mitigation countermeasures. The numerical results show that: (a) The vibration mitigation effect in the first vortex “lock-in” region is not sensitive to the position of the aerodynamic countermeasures, while the vibration mitigation effect in the second “lock-in” region is more sensitive to the position of the aerodynamic countermeasures; (b) The phase of aerodynamic lift and displacement response is close to synchronization in the ascending section of the VIV amplitude, greater than 90° in the descending section, and close to 180° outside the vortex-induced resonance range. The first vertical VIV region is related to the “Karman vortex street” at the trailing edge of the section, while the second vertical VIV region is related to the vortex shedding at the lower edge of the front end of the section and the secondary vortices on the upper surface.

1. Introduction

The composite deck, normally composed of steel girders and concrete slab, is considered to be an extremely competent type of medium-span cable-stayed bridge due to its good material properties and economic cost (Su et al., 2015). Among the structural form of the steel girder and concrete slab, the rib-slab girder is the most economical choice which commonly known as Π -shaped girder section. Nevertheless, because of its bluff body nature, composite bridge decks are prone to suffer from vortex-induced vibration (VIV) at low wind speeds, severe VIVs have occurred on some built composite deck bridges. For example, the Deer Isle-Sedgwick Suspension Bridge with a 2-H-shaped steel edge

girder, which constructed around the same time as the old Tacoma Narrows Bridge, has a history of objectionable VIVs as the wind speed is in the range of 9.11–15.9 m/s (Kumarasena et al., 1991). The Kessock Bridge with two-I-shape steel edge girder (also known as the Π -shaped section) experienced large VIV response three times between October 1991 and May 1992 as the wind speed was approximately around 22 m/s (Owen et al., 1996). Although the VIV of bridge is not as dangerous as flutter instability, the large vibration amplitudes may affect the driving and walking comforts and may even cause local damage and reduce the structural fatigue life (Larsen, 1993; Ge et al., 2022). Since more and more composite cable-stayed bridges with the Π -shaped section deck have been built in recent years, it is essential to study the VIV

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Table 1
Some built Π -shaped bridges.

bridge name	width/m	Height /m	Vertical frequency/Hz	Torsional frequency/Hz	Mass/(kg/m)	Mass moment inertia /(kg·m ² /m)
Yangpu Bridge	30.5	3	0.267	0.517	/	/
Xupu Bridge	35.95	3	0.334	0.666	/	/
Kezhushan Bridge	35	3	0.338	0.716	/	/
Qinglan Bridge	34	3	0.334	0.836	5.01E+04	4.43E+06
Wanguan Bridge	31.5	3	0.318	0.596	3.97E+04	1.16E+06
Malutang Bridge	31.84	3.2	0.303	0.744	5.25E+04	3.99E+06

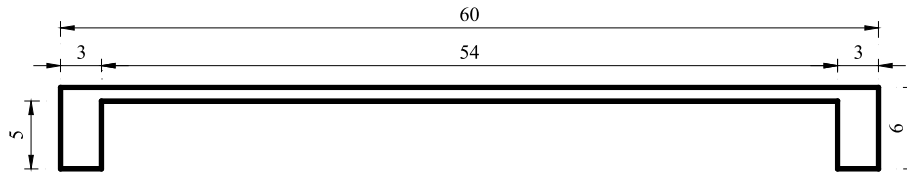


Fig. 1. Geometric of proposed Π -shaped girder section model(Unit: cm).

Table 2
Parameters for tested section model.

Parameter	Model Design Values	Measured value
Width m	0.6	0.6
Height m	0.06	0.06
Vertical frequency Hz	3.36	3.32
Torsional frequency Hz	7.79	7.81
Mass per unit length kg/m	11.12	11.39
Mass moment of inertia per unit length kg·m ² /m	0.3856	0.3806
Vertical damping ratio %	0.5	0.288
Torsional damping ratio %	0.5	0.314

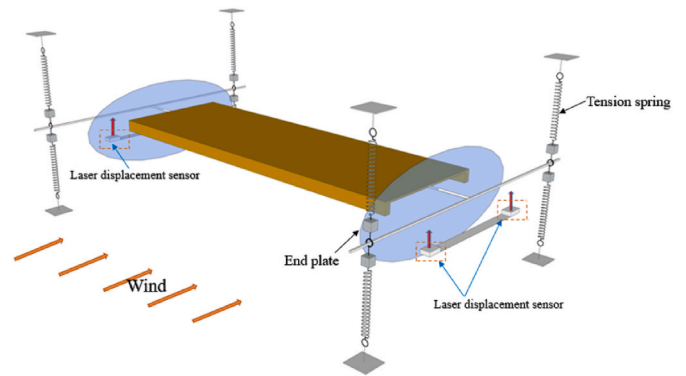


Fig. 3. Schematic diagram of the layout of the laser displacement sensor.

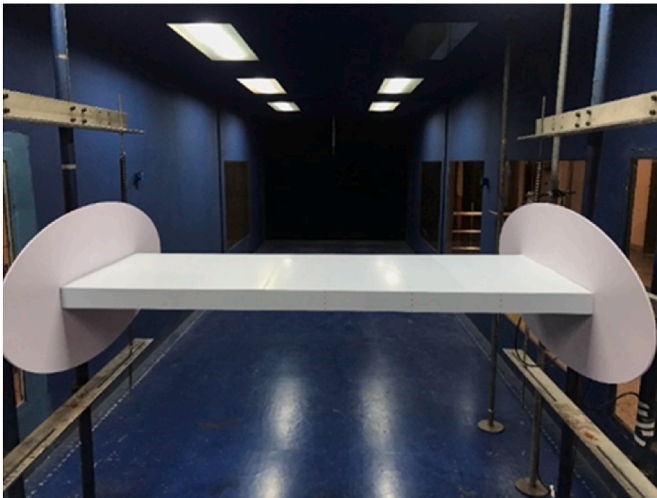


Fig. 2. Experimental vibration section model.

characteristics of long-span bridges with this type decks, hence to investigate the mechanism of different countermeasures and to improve the stability of Π -shaped deck.

The typical characteristic of VIV is that as the frequency of vortex-induced forces, resulted from vortex shedding, approaches the modal frequency of the bluff body, the structure could exhibit a relatively large limit cycle oscillations (LCOs) within a “lock-in” range of wind velocity. The bridge-deck-like bluff bodies with long afterbody normally have several fixed separation points, which vary with Strouhal number.

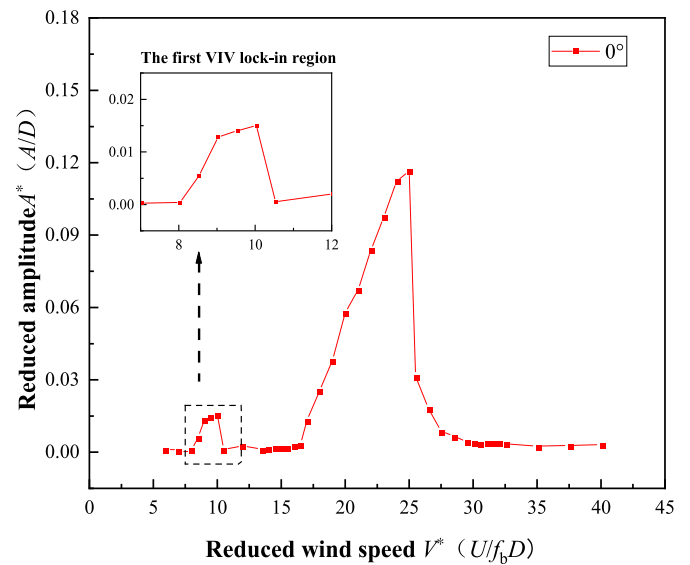


Fig. 4. Vertical VIV response of original section.

However, due to the separation vortex shedding with different Strouhal numbers and “frequency demultiplication” phenomenon, the vertical vibration usually emerges of two “lock-in” regions (Wu and Kareem, 2012). Nakamura and Nakashima (1986) observed the secondary vortex

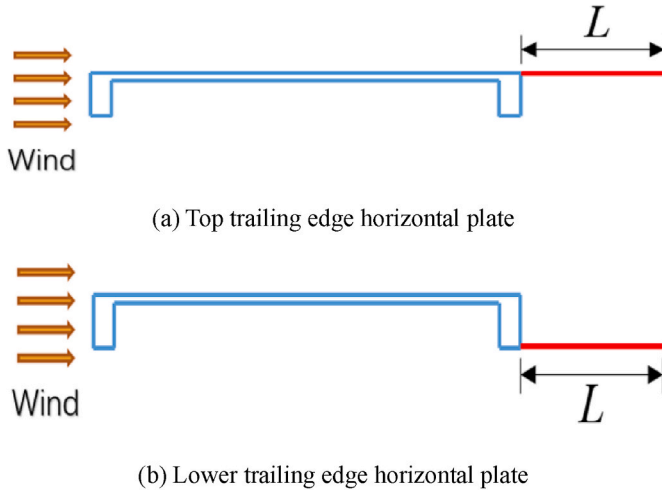


Fig. 5. Schematic diagram of rear edge horizontal plate layout.

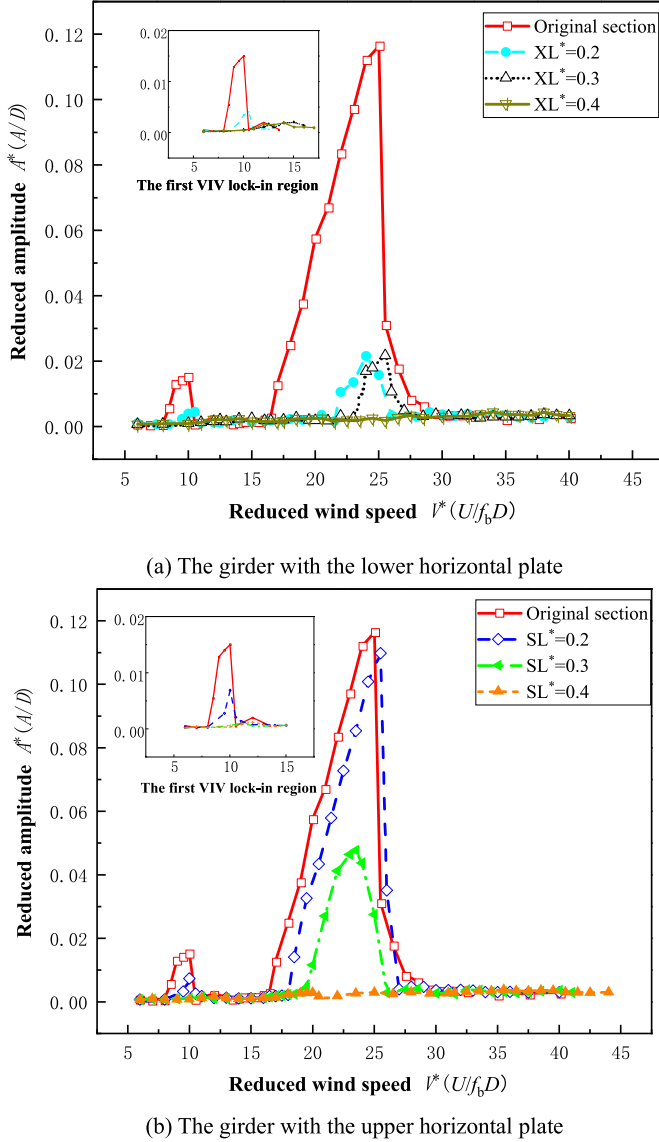


Fig. 6. Vertical VIV response of the girder with horizontal plate.
Notes: XL represents the lower horizontal plate, SL represents the upper horizontal plate.

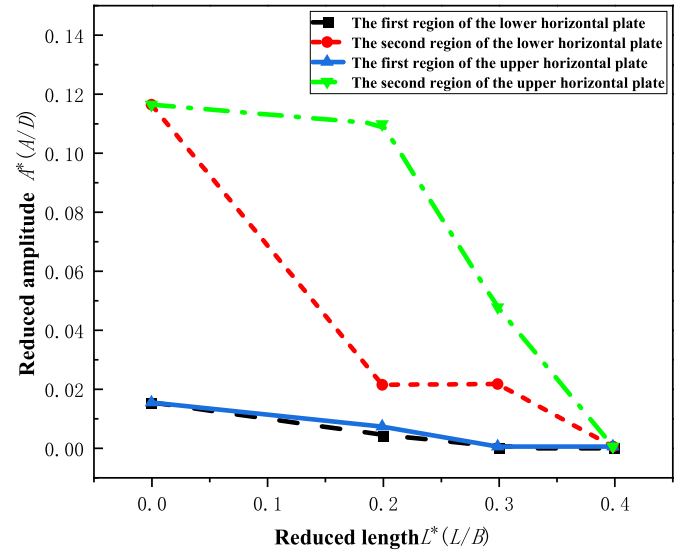


Fig. 7. Change of amplitude extreme value with reduced length.

along with the elongate sharp-edged bluff bodies (e.g., typical bridge decks) since no similar phenomenon was observed for oscillating circular cylinders. Larose et al. (2003) studied the VIV response of Stonecutters bridge twin-box deck based on a 1:20 scaled sectional model subject to different Reynolds numbers. Two distinct “lock-in” regions for vertical VIV were observed in experimental results. Diana et al. (2006) observed VIV of a multi-box bridge deck in detail in the wind tunnel tests of Messina Strait suspension bridge. The experimental results presented a two “lock-in” ranges, which were identified as vertical mode. Noguchi et al. (2020) obtained two “lock-in” regions for vertical VIV through the spring-supported free vibration wind tunnel tests of a box girder sectional model. Hu et al. (2019) applied the experimental method to examine the VIV of a streamlined closed-box girder with handrail, barrier and maintenance trace, the results presented two or three vertical bending vortex vibration intervals at initial attack angle of $+3^\circ$. Besides, the two vertical VIV “lock-in” regions were also observed and studied by Zhang et al. (2020) and Bai et al. (2020) according to the wind tunnel tests. The aforementioned researches mainly focus on the VIV characteristics in the second “lock-in” region with high wind speed, less attention was paid to the region with low wind speed which have higher probability to occur VIV with relatively small amplitude observed in real bridges. Besides, it is also necessary to avoid large-amplitude VIV in the “lock-in” region with high wind speed in the design process.

In the application of bridge engineering, aerodynamic countermeasures are widely applied on mitigating the bridge VIV hence improving wind resistance. Kubo et al. (2001) investigated the aerodynamic performances of I-shape deck subject to different intervals between the two I-shape girders. Flow visualization and unsteady pressure measured technologies were employed to study the mechanism of VIV response mitigated by decreasing the girders interval. Besides, fairing is a common measure to improve the VIV of cable-stayed bridges with typical I-shape deck, which was first applied and studied by Wardlaw (1972). Sakai et al. (1993) utilized wind tunnel tests and computational flow analyses to explore the effects of different attachments, aimed to develop a simple, economical and effective aerodynamic countermeasures for I-shape deck to mitigate VIV response. In addition, central stabilizing plate and vertical baffle plates installed under the deck were also developed and investigated regarding their efficiency on suppressing the VIV (Irwin, 2008; Li et al., 2018a; Ma et al., 2018). Bai et al. (2020) studied the effects of the sealed traffic barrier countermeasure to improving the flutter and VIV stability of bridge through wind tunnel tests for three distinctive real bridge projects. Their results indicated that traffic barrier sealed by regular intervals along the bridge can

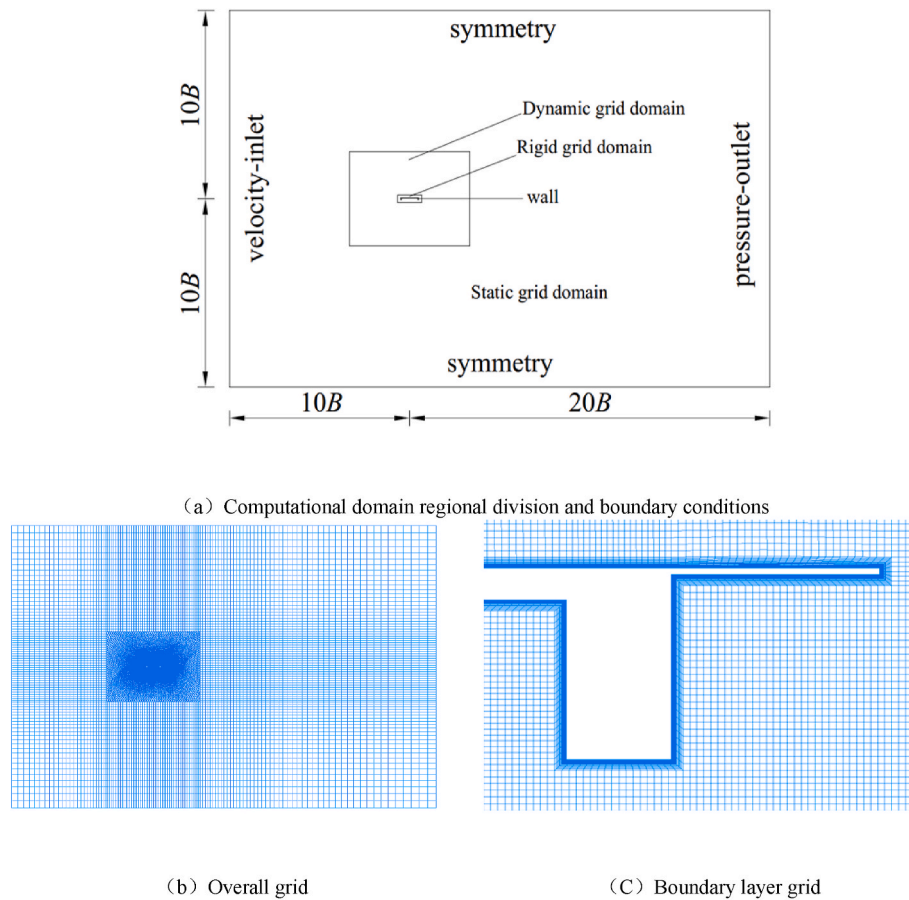


Fig. 8. Computational grids.

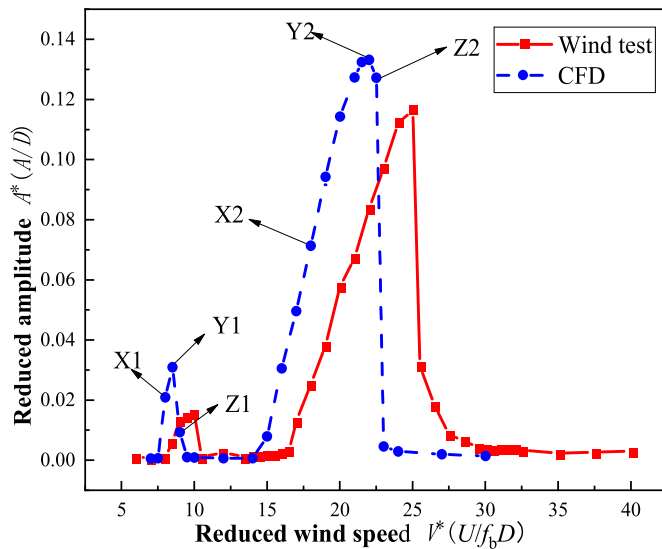


Fig. 9. The original section VIV response obtained by numerical simulation and wind tunnel test.

effectively mitigate the VIV. With the emerging growth in computational capability, the computational fluid dynamics (CFD) based numerical simulation method is becoming a powerful tool for the investigation of the wind-induced vibration of bridges. Hence, it is a promising approach to explore the VIV triggering mechanism and propose different aerodynamic countermeasures by simulating the flow

pattern around the bridge deck (Sarwar and Ishihara, 2010; Vaz et al., 2016; Li et al., 2018b; Ma et al., 2018; Zhang et al., 2020; Bai et al., 2021). Most of the previous researches utilized the CFD approach to simulate the vortex pattern in the wake of the flow after the steady bridge deck. However, the vibrating bridge could have the interaction with the flow around the structure, thus need the fluid-structure interaction (FSI) simulation based on the dynamic mesh technique to deeply investigate the mechanism of the VIV in two different "lock-in" regions.

At present, numerous studies related to the VIV performance of long-span bridges with composite deck have been conducted, however few of them were presented to investigate the characteristics of two vertical VIV "lock-in" regions and effects of vibration mitigation countermeasures on each regions. In this study, a typical Π -shaped deck with two vertical VIV "lock-in" regions is employed to explore the VIV mechanism and examine several mitigation countermeasures. First, the VIV response of the original section and those of the sections with different lengths and positions of horizontal plates are experimentally investigated through the section model wind tunnel tests. Thereafter, CFD approach with dynamic mesh technique is applied to simulate the flow field around the original section with different wind speeds, and the flow field around the section with aerodynamic countermeasures as the wind speed at the amplitude extreme point of two vertical VIV "lock-in" regions. Finally, based on the simulation results, such as the flow pattern around the bridge decks and aerodynamic forces in the "lock-in" regions, the mechanism of the aerodynamic countermeasures in suppressing the vibration and effects on each "lock-in" regions are systematically explained. The concluding from the present study highlighted the important effects of two "lock-in" VIV on Π -section girder, and hence it is beneficial to find aerodynamic measures for Π -section girder to suppress VIV efficiently.

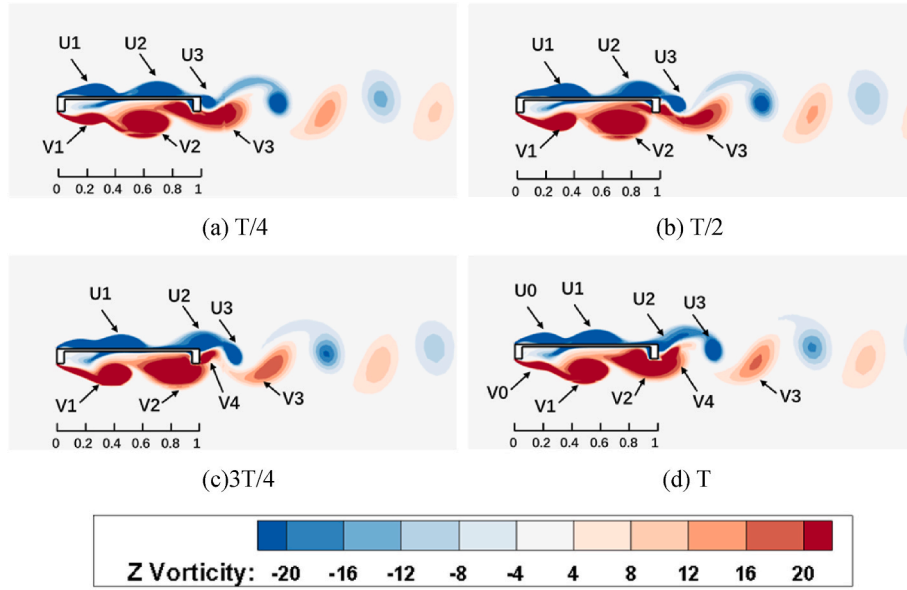


Fig. 10. Vortex evolution diagram in vibration period T at Y1.

2. Experimental study of vortex-induced vibration

2.1. Experimental setup

Table 1 summarizes the key parameters of some built Π -shaped cable-stayed bridges. In this study, the geometric and structure dynamic characteristics of the section model employed in wind tunnel test are determined by referring to the built bridges in Table 1. It is finally determined that the actual size of the Π -shaped girder is 3m high and 30m wide. The aspect ratio of the main girder is $B/D = 10$. The geometric scale ratio is 1:50, the design section of the model is shown in Fig. 1, the specific design parameters and measured parameters are shown in Table 2.

The section model wind tunnel test is carried out through a closed-circuit wind tunnel, and the experiments under 0° angle of attack of original section and those with aerodynamic countermeasures are carried out in a uniform flow field. The section model test is shown in Fig. 2. During the test, the laser displacement sensor was used to record the vibration signal. Three Laser displacement sensors were set up in the test, and the arrangement positions are shown in Fig. 3.

2.2. Vortex-induced vibration response of the original section

In the test of the original section without any aerodynamic countermeasures, the vertical VIV response with two “lock-in” regions is observed, shown in Fig. 4. In the figure, the abscissa represents the reduced wind speed $V^* = U/f_b D$, and the ordinate represents the reduced amplitude $A^* = A/D$, where U represents the incoming wind speed, D represents the characteristic height of the main girder, A represents the model amplitude, and f_b represents the vertical frequency of the section model. It can be seen from Fig. 4 that the reduced wind speed range in the first VIV “lock-in” region is 8.03~10.54 with the maximum amplitude of $A^*_{max} = 0.0149$ occurs at $V^* = 10.04$; the reduced wind speed range in the second VIV “lock-in” region is of 16.57~29.62, and the maximum amplitude of $A^*_{max} = 0.1163$ occurs at $V^* = 25.1$.

Shiraishi and Matsumoto (1983) has summed up the calculation formula of the vortex-induced vibration onset reduced wind speed through a large number of experiments. According to this formula, it can be obtained:

The first vertical VIV region: $V_{cr} = (1/n) \times 1.67 \times (B/D) = (1/2) \times 1.67 \times (60/6) = 8.3$

The second vertical VIV region: $V_{cr} = (1/n) \times 1.67 \times (B/D) = (1/1) \times 1.67 \times (60/6) = 16.7$

In the experiment, the onset reduced wind speed $V_{cr} = 8.03$ in the first vertical VIV region; the onset reduced wind speed $V_{cr} = 16.57$ in the second VIV region, it can be seen that the calculated results by the formula are in good agreement with the experimental results.

2.3. Vortex-induced vibration response with aerodynamic countermeasures

Fig. 5 demonstrate the schematic diagram of setting a horizontal plate at the rear of the downstream side box of the Π -shaped girder. The length of the horizontal plate is dimensionless, and the reduced length $L^* = L/B$ is defined, where L represents the length of the horizontal plate, and B is the characteristic width of beam.

Fig. 6 is the response diagram of the vertical VIV of the main girder under seven working conditions with different horizontal plate locations and lengths (X represent lower trailing edge position; S represent top trailing edge position). It can be seen from Fig. 6 that when $XL^* = 0.2$, the amplitude of the first VIV region is 73.3% lower than that of the original section, and the amplitude of the second VIV region is 81.9% lower than that of the original section; as $XL^* = 0.3$, the first VIV region disappears, and the amplitude of the second VIV region decreases by 81.7% compared with the original section amplitude; as $XL^* = 0.4$, the two VIV regions disappears. When $SL^* = 0.2$, the amplitude of the first VIV region is 54.7% lower than that of the original section, and the amplitude of the second VIV region is 5.7% lower than that of the original section; as $SL^* = 0.3$, the VIV region disappears, and the amplitude of the second VIV region decreases by 59.3% compared with the original section amplitude; as $SL^* = 0.4$, the two VIV regions disappears. Fig. 7 shows the relationship between the extreme value of the reduced amplitude and the reduced length of the horizontal plate. From the analysis of this figure, it can be seen that the amplitude reduction trend of the upper and lower horizontal plates with the same reduced length in the first VIV region is consistent; the amplitude reduction of the lower trailing edge horizontal plate in the second vortex vibration interval is significantly better than that of the upper trailing edge horizontal plate with the same reduced length. It can be seen from Figs. 6 and 7 that the vibration reduction effect of the lower horizontal plate is better than that of the upper horizontal plate under the condition of the same reduced length.

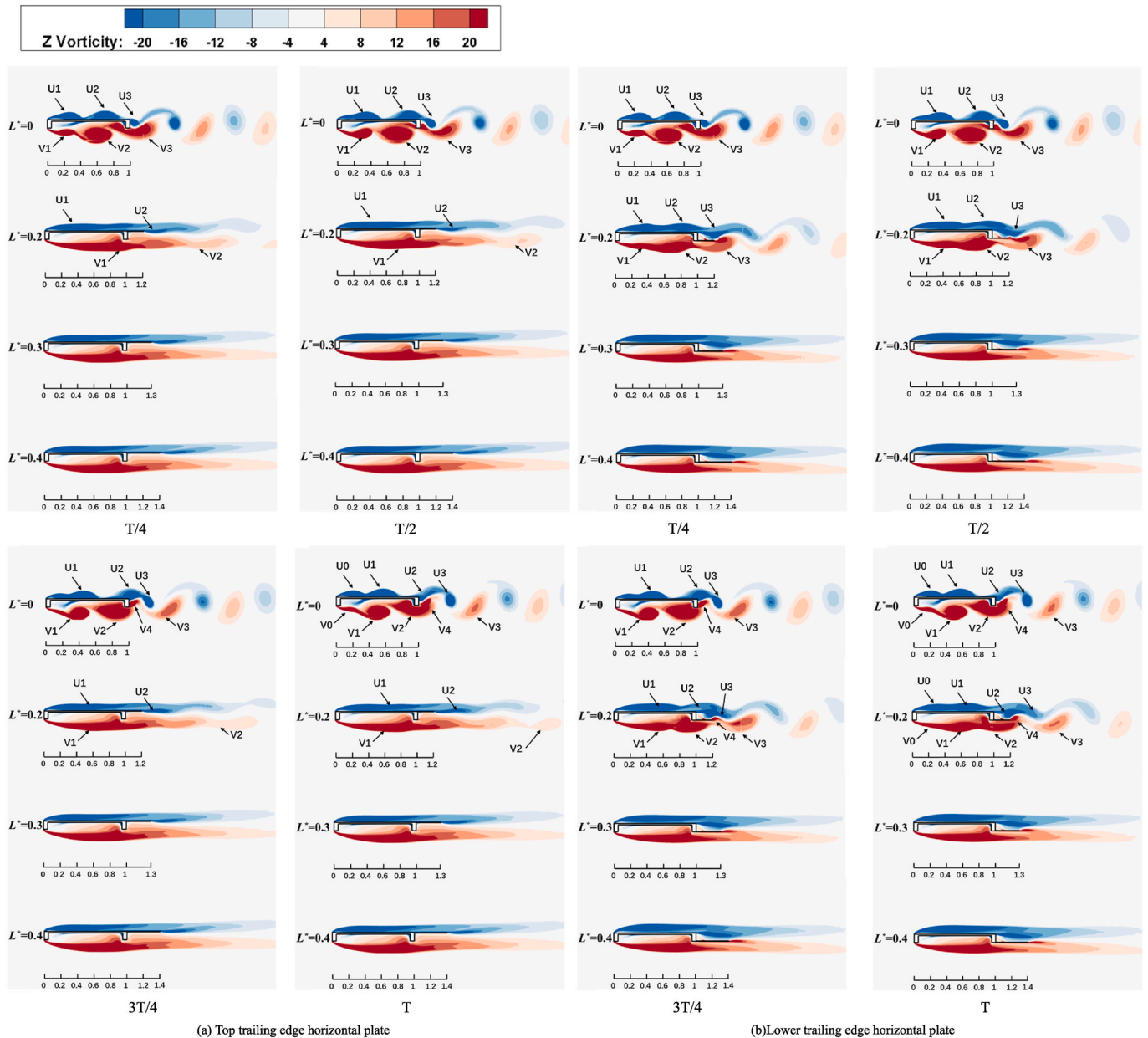


Fig. 11. Vortex evolution diagram of different horizontal plate sections of Y1.

3. Numerical simulation of vortex-induced vibration

3.1. Numerical simulation setup

CFD numerical simulation is implemented using the FLUENT software. The grid used for the calculation is shown in Fig. 8. Fig. 8(a) shows that the mesh is composed of rigid mesh area, dynamic mesh area, and static mesh area and computational domain boundary conditions; the inlet of the computational domain adopts the velocity-inlet boundary condition, the outlet adopts the pressure-outlet boundary condition, the upper and lower boundaries adopt the no-slip symmetry boundary, and the surface of the main beam adopts the wall boundary condition. When the inlet wind speed $U = 4.3$ m/s, the Reynolds number $Re = 14418$. Fig. 8(b) is a schematic diagram of the overall grid, the total number of grids is 92289; Fig. 8(c) shows the boundary layer of the grid, the first layer of the boundary layer grid is 6×10^{-5} m, and the boundary layer has a total of 30 layers. The calculation adopts the SST $k-\omega$ turbulence model, the solution method adopts the SIMPLE second-order implicit

algorithm, the time step is set to 0.0005s, and the convergence residual is controlled at 10^{-5} .

Due to the movement of the structure, the fluid boundary conditions will change, which will disturb the boundary layer and affect the shedding of the vortex (Rockwell and Naudascher, 1978; Shiraishi and Matsumoto, 1983). In order to more reasonably simulate the flow field structure when the structure generates vortex-induced vibration. The dynamic grid technology is used in the calculation, which can realize the simulation of the flow domain changing with time. The basic calculation procedure: the first step is to divide the grid and then steady-state calculation under the static state of the main girder. In the second step, the steady-state calculation is transferred to the transient calculation after the convergence residual is stable. The third step is to release the UDF after the flow field of the transient calculation is stable, realize the grid update, let the main girder move under the action of aerodynamic force, and achieve the fluid-solid coupling. Finally, the calculation ends when the main girder motion is judged to be stable through the vibration response time history curve.

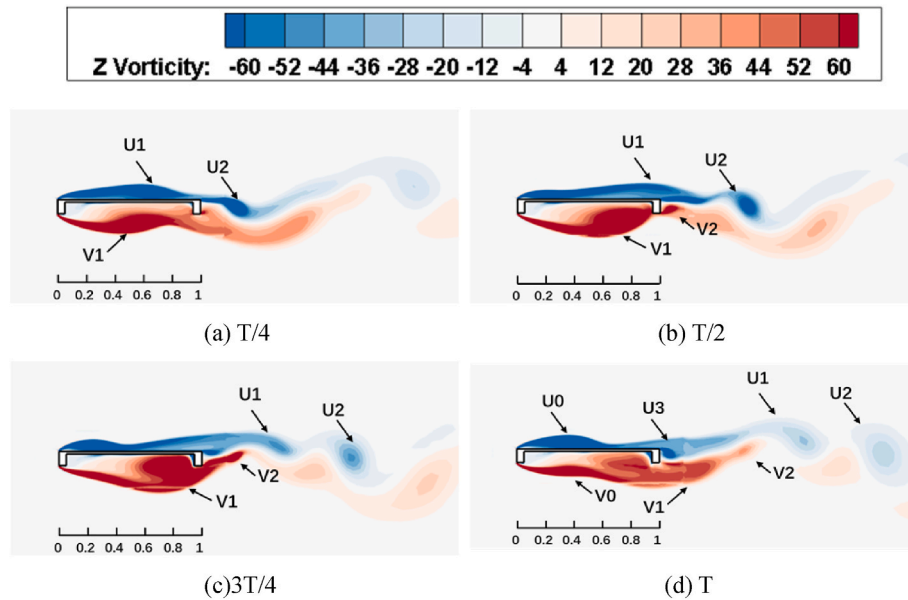


Fig. 12. Vortex evolution diagram in vibration period T at Y2.

3.2. Numerical simulation response results

The numerical simulation of VIV of the main girder is carried out every 0.2 m/s wind speed in the range of 1.4 m/s~6 m/s wind speed, and the vortex vibration range is properly encrypted to determine the starting wind speed, the ending wind speed, and the wind speed at the extreme point of the amplitude. A total of 23 calculations were performed at different wind speeds. The vortex vibration response of the original section obtained by CFD numerical simulation and wind tunnel test is shown in Fig. 9. It can be found from Fig. 9 that the numerical simulation results are different from those of the wind tunnel test, but the variation law of the amplitude with the wind speed is basically the same, and there is also a two lock-in regions of vertical VIV. In the numerical simulation results, the rising point, extreme point and falling point of the first locking region VIV are defined as X1, Y1, Z1, and the corresponding reduced wind speeds are 8.03, 8.53, 9.03 respectively; similarly, the second locking region is X2, Y2, Z2, the corresponding reduced wind speeds are 18.07, 22.08, 22.59, respectively. The numerical simulation of the section with the horizontal plate under the wind speed at the extreme point of the amplitude is carried out to obtain the flow field information of different sections.

CFD numerical simulation shows that the reduced wind speed range in the first VIV “lock-in” region is 7.53~9.54, and the maximum amplitude $A^*_{max}=0.0304$ appears in $V^*=8.53$; the reduced wind speed range in the second VIV lock-in region is 14.06~22.08, and the maximum amplitude $A^*_{max}=0.1333$ appears in $V^*=22.08$. The wind speed error at the amplitude extreme points of the two vortex vibration regions is 15%.

4. Characteristics of vertical vortex-induced vibration in two “lock-in” regions

4.1. Vortex shedding in the first “lock-in” region

Fig. 10 shows the vorticity evolution process of the main girder vibration in one cycle T at the wind speed Y1 at the amplitude extreme point in the first VIV lock-in region of the original section. During a cycle of time when the airflow flows through the main girder section, there is a “Karman vortex street” at the tail of the main girder. In one period, vortices U1, U2, U3 are generated in the upper part of the main girder, V1, V2, V3 and vortex V4 at the beginning of the next cycle are

generated in the lower part of the main girder. The vortices U3 and V3 successively separated from the upper and lower surfaces of the main girder and evolved into the initial stage of the “Karman vortex street” in the wake, and finally formed a stable “Karman vortex street” at the tail. The above-mentioned V0, V1, V2, V3 and V4 represent the development process of the separation vortex at the lower edge of the section; U0, U1, U2, and U3 represent the development process of the separation vortex at the upper edge of the section. The vortex defined hereinafter also have the above meaning.

Fig. 11 shows the vorticity evolution process of the main girder vibration in one period T and the comparison of the vorticity maps at the same time when the horizontal plates with different reduced lengths are used at the wind speed of Y1. It can be seen from Figs. 7 and 11 that the first VIV lock-in region disappears when the horizontal plate $L^*\geq 0.3$. The reason is that the flow on the upper and lower part of the section is stable, and compared with the original section, the horizontal plate is installed so that there is no “Karman vortex street” at the trailing edge of the section. Fig. 11(a) shows that when $L^*=0.2$ upper horizontal plate is used, the V1 vortex does not develop and separate into an independent vortex at the lower part of the section, and there is only a weak vortex V2 separation and no obvious “Karman vortex street” in the wake; The morphology of the upper U1 vortex is also significantly different from the original section. It can be seen from Fig. 11(b) that when $L^*=0.2$ lower horizontal plate is used, there is no obvious dividing point between the V1 and V2 vortices at the lower part of the section, and the area of the V3 vortex vorticity greater than 20 is also smaller than the original section, which indicates that the V3 vortex strength decreases; the upper U3 vortex separation also produces hysteresis and intensity reduction; the wake “Karman vortex street” vortex intensity is also smaller than the original section. This indicates that the horizontal plate at different positions suppresses the VIV by suppressing the development of vortices in the upper and lower sections of the section, thereby changing the “Karman vortex street” in the wake. It is revealed that the first VIV lock-in region is caused by the “Karman vortex street” in the wake.

4.2. Vortex shedding in the second “lock-in” region

Fig. 12 shows the vorticity evolution process of the main girder vibration in one cycle T at the wind speed Y2 at the amplitude extreme point in the second VIV lock-in region of the original section. During a

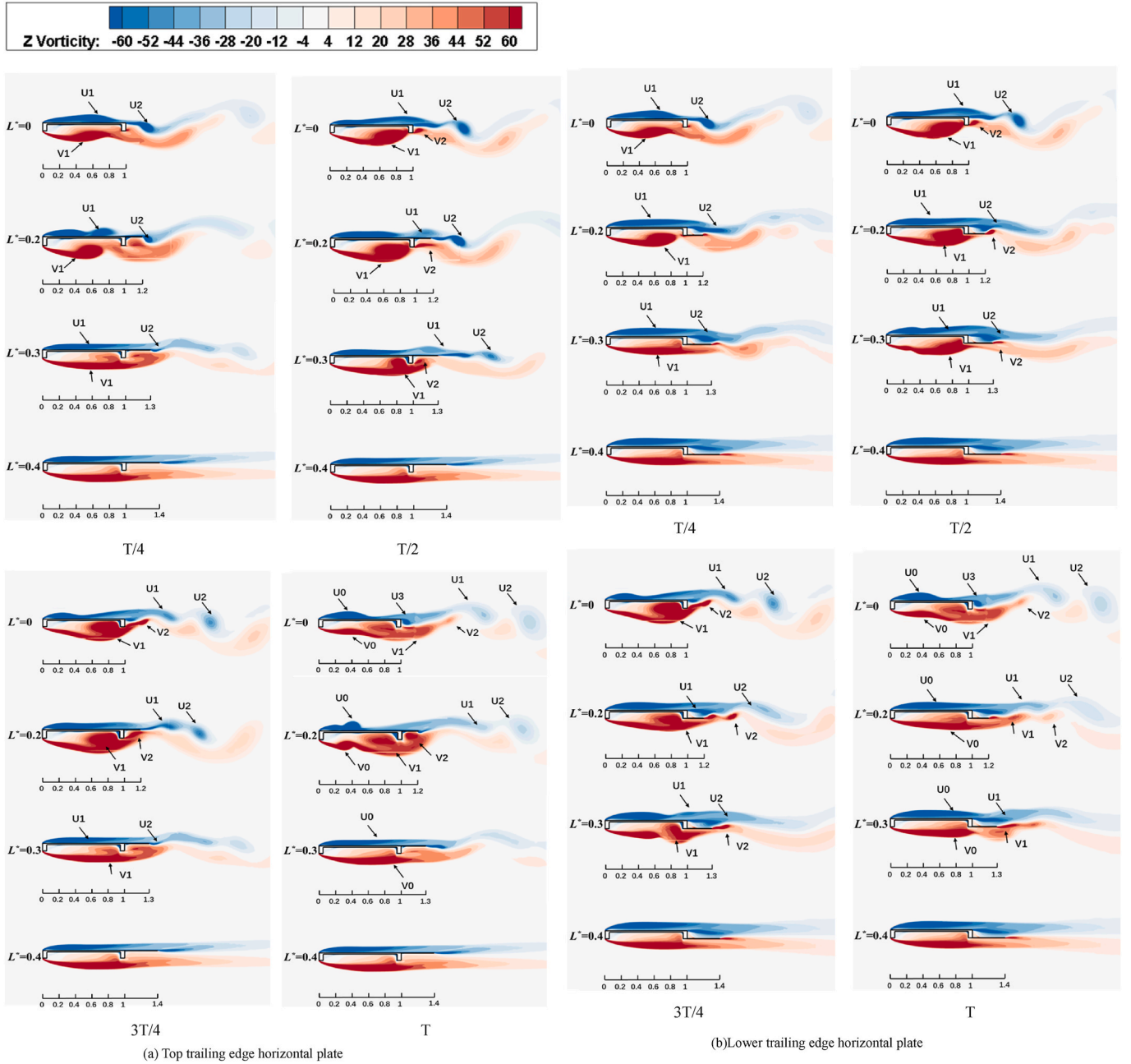


Fig. 13. Vortex evolution diagram of different separation plate sections of Y1.

cycle of time when the airflow flows through the main beam section, vortices U1 and U2 are mainly generated in the upper part of the main girder; V1 and V2 are generated in the lower part of the main girder. It can be seen from the whole cycle that the velocity of the upper vortex of the main girder is faster than that of the lower vortex and the scale of the upper vortex is significantly smaller than that of the lower vortex, and the upper vortex will form an obvious shedding vortex (U2) after it leaves the main girder. The vortex generated by the separation of the lower edge of the front end of the main girder develops into a huge vortex equivalent to the half width of the main girder.

Fig. 13 shows the vorticity evolution process of the main girder vibration in one period T and the comparison of the vorticity maps at the same time when the horizontal plates with different reduced lengths are used at the Y2 wind speed. It can be seen from Figs. 7 and 13 that the second VIV lock-in region disappears when the horizontal plate $L^* = 0.4$. This is because the flow is stable in the upper and lower parts of the

section. Compared with the original section, the horizontal plate is installed so that there is no vortex generation in the lower section and vortex shedding in the upper section. It can be seen from Fig. 13(a) that when $L^* = 0.2$ upper horizontal plate is used, compared with the original section, the size and intensity of the upper separation vortices U1 and U2 became smaller during the development process, for example the scale of the U2 is about $0.1B$ while those under the original section is about $0.2B$ at $T/4$; the development process of the lower vortex V1 was similar to the original section, the horizontal scale of the V1 is similar to the original one, which are both $0.7B$ at $T/4$ as an example. It indicates that the development of the vortex in the upper part of the section is inhibited. Fig. 7 shows that the extreme value of the VIV amplitude in the second VIV “lock-in” region of the upper horizontal plate (the green dash-dotted line with downward-pointing triangle marker) at reduced length (L^*) of 0.2 is about 5.7% lower than that of the original section (the green dot dash line with downward-pointing triangle marker at

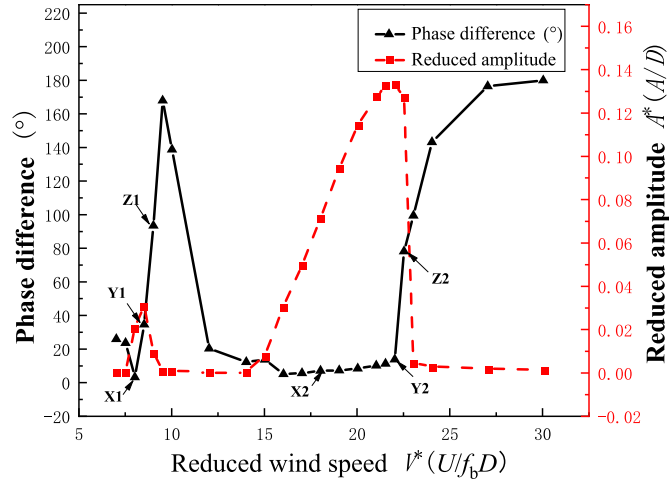


Fig. 14. Phase difference between aerodynamic lift and displacement response.

reduced length of 0). It can be seen from Fig. 13(a) that when $L^* = 0.3$ upper horizontal plate is used, compared with the original section, the area of the -60 vorticity (the deep blue vortex area) of vortices U1 and

U2 at each process is significantly reduced; The shape and scale of the lower vortex V1 at different times is significantly different from the original section, for example, the horizontal scale is about $1.3B$, which is about two times of the original section at $T/4$; meanwhile, the area with vorticity greater than 60 at $T/2$, $3T/4$ and T times are also significantly reduced. This indicates that the development of V1 vortex is more obviously inhibited in the lower part of the section. In Fig. 7, the results shows that the extreme VIV amplitude in the second VIV “lock-in” region of the upper horizontal plate (the green dash-dotted line with downward-pointing triangle marker) at reduced length of 0.3 is about 59.3% lower than that of the original section (the green dot dash line with downward-pointing triangle marker at reduced length of 0). It can be seen from Fig. 13(b) that when $L^* = 0.2$ lower horizontal plate is used, at $T/4$ and $T/2$, the horizontal scale of the lower vortex V1 is similar to the original one, which are both $0.7B$ at $T/4$ as an example. However, at $4T/3$ and T , the area of the V1 vorticity greater than 60 is smaller than the original section, especially at T process. It is shown that the V1 vortex is inhibited from developing in the lower part of the section. It could be noticed in Fig. 7, the extreme VIV amplitude value in the second VIV “lock-in” region of the lower horizontal plate (the red dash line with circle marker) at reduced length of 0.2 is about 81.9% lower than that of the original section (the red dash line with circle marker at reduced length of 0). It can be seen from Fig. 13(b) that when

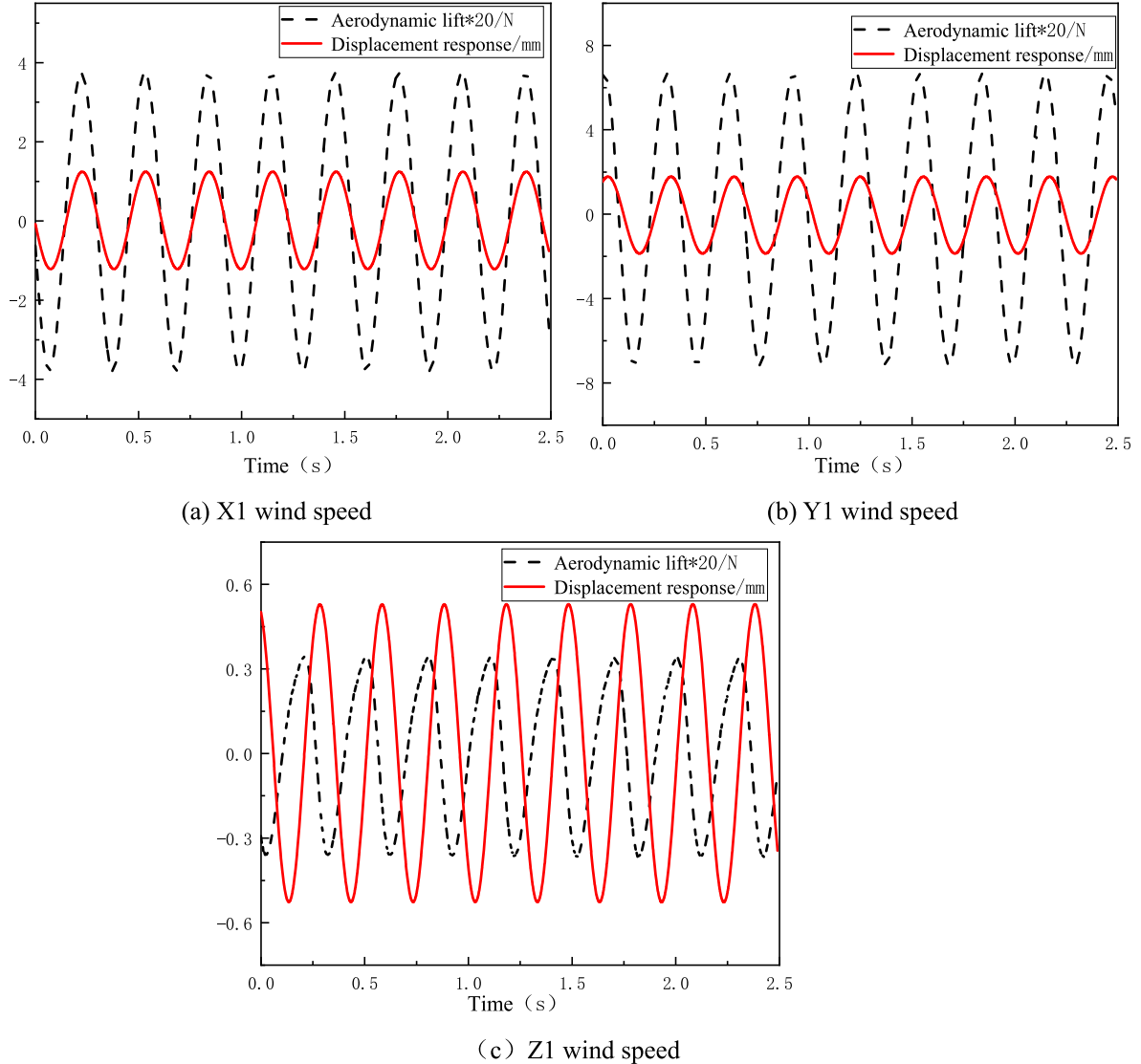


Fig. 15. Comparison of the phase difference between aerodynamic lift and displacement response in the “lock-in” region.

$L^* = 0.3$ lower horizontal plate is used, the shape and scale of the lower vortex V1 are significant differ from the original section at each time process, for example the horizontal scale is about 1B at $T/4$, while those of the original section is about 0.7B; meanwhile, the area of the vorticity larger than 60 (the deep red area) of the V1 vortex at the section opening side is much smaller than that of original section at the time of $3T/4$. All these indicate that the development of V1 vortex is severely inhibited in the lower part of the section. In Fig. 7, the extreme VIV amplitude value in the second VIV “lock-in” region of the lower horizontal plate (the red dash line with circle marker) at reduced length of 0.3 is about 81.7% lower than that of the original section (the red dash line with circle marker at reduced length of 0). Through the above analysis, it can be seen that the secondary vortex generated by the movement of the structure and the separation vortex at the lower part of the structure are the causes of the second VIV lock-in region, and the separation vortex at the lower part of the structure is the main reason for the second VIV lock-in region.

4.3. Aerodynamic lift forces in “lock-in” region

During the numerical simulation, the aerodynamic forces of the main girder is monitored, and the aerodynamic lift plays a major role in the vertical VIV (Wu and Laima, 2021). Fig. 14 shows the phase difference between the aerodynamic lift and the displacement response of the main girder obtained by numerical calculation. The phase difference is equal to the absolute value of the main frequency phase subtraction obtained by Fourier transforming the aerodynamic lift and displacement time-history curves. It can be seen from Fig. 14 that there is a phase difference between the aerodynamic lift and the displacement response, and it changes with the VIV lock-in region. The reduced wind speed at the minimum phase difference in the two VIV lock-in regions is in the vortex vibration rising interval. The phase difference gradually increases in the rising interval of VIV and phase difference at the peak point of the vortex vibration has a significant increase compared with the phase difference in the rising interval. The phase difference in the interval without vortex vibration is close to 180° , and the aerodynamic lift suppresses the vibration of the main girder.

Taking only the first lock-in region as an example, Fig. 15 shows the corresponding relationship between the displacement response time history and the aerodynamic time history at the three key reduced wind speed points X1, Y1, and Z1. It can be seen from Fig. 15 that the amplitudes of displacement and aerodynamic lift first increase and then decrease in the lock-in region. The phase difference increases from 2.4° at X1 point to 33.8° at Y1 point, indicating that the phase difference is small in the rising interval of VIV, and the aerodynamic lift provides energy for the model vibration. The phase difference increases from 33.8° at Y1 point to 92.97° at Z1 point, indicating that the phase difference increases significantly in the descending interval of VIV. In the final stage, the aerodynamic force is opposite to the displacement direction, therefor the main girder vortex vibration is suppressed by aerodynamic lift.

5. Concluding remarks

In this paper, the wind tunnel test and numerical simulation are carried out for the Π -shaped girder widely used in long-span bridges. The characteristics of two “lock-in” regions for vertical vortex-induced vibration of Π -shaped girder are studied, and the following results are obtained:

- (1) The comparison between the wind tunnel test and the numerical simulation results shows that the influence mechanism of the upper and lower horizontal plates on the wake is the same in the first VIV lock-in region, so it is not sensitive to the position of the horizontal plates. In the second vortex vibration interval, the upper horizontal plate mainly changes the upper flow field

structure of the main girder, and the lower horizontal plate mainly changes the lower flow field structure, but the vibration suppression effect of the lower horizontal plate is better than that of the upper horizontal plate. Therefore, the aerodynamic countermeasures that can change the flow field at the lower part of Π -shaped girder should be adopted.

- (2) There is always a phase difference between the aerodynamic lift and the displacement response. Analysis of the phase difference in the two “lock-in” regions shows that the minimum phase difference (less than 5°) appears in the rising stage of the vortex vibration, and the aerodynamic lift promotes the growth of the vibration amplitude of the girder. The phase differences are all greater than 90° in the vortex vibration descending stage, and the aerodynamic lift suppresses the VIV of the main girder. The final phase difference approaches 180° outside the “lock-in” region.
- (3) The vortex shedding shape obtained by numerical simulation shows that the first VIV “lock-in” region is caused by the “Karman vortex street” formed by the periodic shedding of vortices at the trailing edge of the section; and VIV can be suppressed by setting the horizontal plate to change the vortex strength of the “Karman vortex street” in the wake. The vortex shedding and wake in the second VIV “lock-in” region are significantly different from the “Karman vortex street” of the first VIV “lock-in” region. The secondary vortices generated on the upper surface due to structural motion and the shedding vortices generated at the lower edge of the front end is the main reason for the second VIV “lock-in” region; and VIV can be suppressed by setting the horizontal plate to reduce the vortex size at the lower part of the main girder.

CRediT authorship contribution statement

Jiawu Li: Conceptualization, Methodology, Resources, Supervision, Writing – review & editing. **Peng Wu:** Methodology, Investigation, Writing – original draft. **Jianming Hao:** Conceptualization, Software, Supervision, Formal analysis, Writing – original draft, review. **Hui Pan:** Validation, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- Bai, H., Ji, N., Xu, G.J., Li, J., 2020. An alternative aerodynamic mitigation measure for improving bridge flutter and vortex induced vibration (VIV) stability: sealed traffic barrier. *J. Wind Eng. Ind. Aerod.* 206, 104302.
- Bai, H., Li, R., Xu, G.J., Kareem, A., 2021. Aerodynamic performance of Π -shaped composite deck cable-stayed bridges including VIV mitigation. *J. Wind Eng. Ind. Aerod.* 208, 104451.
- Diana, G., Resta, F., Belloli, M., Rocchi, D., 2006. On the vortex shedding forcing on suspension bridge deck. *J. Wind Eng. Ind. Aerod.* 94 (5), 341–363.

- Ge, Y.J., Zhao, L., Cao, J.X., 2022. Case study of vortex-induced vibration and mitigation mechanism for a long-span suspension bridge. *J. Wind Eng. Ind. Aerod.* 220, 104866.
- Hu, C., Zhao, L., Ge, Y., 2019. Mechanism of suppression of vortex-induced vibrations of a streamlined closed-box girder using additional small-scale components. *J. Wind Eng. Ind. Aerod.* 189, 314–331.
- Irwin, P.A., 2008. Bluff body aerodynamics in wind engineering. *J. Wind Eng. Ind. Aerod.* 96 (67), 701–712.
- Kubo, Y., Sadashima, K., Yamaguchi, E., Kato, K., Okamoto, Y., Koga, T., 2001. Improvement of aeroelastic instability of shallow π section. *J. Wind Eng. Ind. Aerod.* 89 (14–15), 1445–1457.
- Kumarasena, T., Scanlan, R.H., Ehsan, F., 1991. Wind-induced motions of deer isle bridge. *J. Struct. Eng.* 117 (11), 3356–3374.
- Larose, G.L., Larsen, S.V., Larsen, A., Hui, M., Jensen, A.G., 2003. Sectional model experiments at high Reynolds number for the deck of a 1018m span cable-stayed bridge. In: *Proceedings of 11th International Conference on Wind Engineering*. Lubbock, pp. 373–380.
- Larsen, A., 1993. Aerodynamic aspects of the final design of the 1624 m suspension bridge across the Great Belt. *J. Wind Eng. Ind. Aerod.* 48 (2), 261–285.
- Li, H., He, X.H., Wang, H.F., Liu, M.T., Peng, S., 2018a. Wind tunnel tests for VIV control measures of a super high cable-stayed bridge with π -cross section. *J. Vib. Shock* 37 (7), 62–68 (in Chinese).
- Li, Y.L., Chen, X.Y., Yu, C.J., Togbenou, K., Wang, B., Zhu, L.D., 2018b. Effects of wind fairing angle on aerodynamic characteristics and dynamic responses of a streamlined trapezoidal box girder. *J. Wind Eng. Ind. Aerod.* 177, 69–78.
- Ma, C.M., Wang, J.X., Li, Q.S., Qin, H., Liao, H.L., 2018. Vortex-induced vibration performance and suppression mechanism for a long suspension bridge with wide twin-box girder. *J. Struct. Eng.* 144 (11), 04018202.
- Nakamura, Y., Nakashima, M., 1986. Vortex excitation of prisms with elongated rectangular, H and T cross-sections. *J. Fluid Mech.* 163, 149–169.
- Noguchi, K., Ito, Y., Yagi, T., 2020. Numerical evaluation of vortex-induced vibration amplitude of a box girder bridge using forced oscillation method. *J. Wind Eng. Ind. Aerod.* 196, 104029.
- Owen, J.S., Vann, A.M., Davies, J.P., Blakeborough, A., 1996. The prototype testing of Kessock Bridge: response to vortex shedding. *J. Wind Eng. Ind. Aerod.* 60, 91–108.
- Rockwell, D., Naudascher, E., 1978. Review- Self-sustaining oscillations of flow past cavities. *Trans. ASME J. Fluid. Eng.* 100, 152–165.
- Sakai, Y., Ogawa, K., Shimodoi, H., Saitoh, T., 1993. An experimental study on aerodynamic improvements for edge girder bridges. *J. Wind Eng. Ind. Aerod.* 49, 459–466.
- Sarwar, M.W., Ishihara, T., 2010. Numerical study on suppression of vortex-induced vibrations of box girder bridge section by aerodynamic countermeasures. *J. Wind Eng. Ind. Aerod.* 98 (12), 701–711.
- Shiraishi, N., Matsumoto, M., 1983. On classification of vortex-induced oscillation and its application for bridge structures. *J. Wind Eng. Ind. Aerod.* 14 (1–3), 419–430.
- Su, Q., Yang, G., Bradford, M.A., 2015. Behavior of a continuous composite box girder with a prefabricated prestressed-concrete slab in its hogging-moment region. *J. Bridge Eng.* 20 (8), B4014004.
- Vaz, D.C., Almeida, R.A., Didier, E., Urgueira, A.P., Borges, A.J., 2016. Improving the aerodynamic performance of Vila-Real Bridge deck-section. *J. Wind Eng. Ind. Aerod.* 156, 72–83.
- Wardlaw, R.L., 1972. *Some Approaches for Improving the Aerodynamic Stability of Bridge Road Decks*. National Research Council Press, Canada.
- Wu, T., Kareem, A., 2012. An overview of vortex-induced vibration (VIV) of bridge decks. *Front. Struct. Civ. Eng.* 6, 335–347.
- Wu, B.C., Laima, S.J., 2021. Experimental study on characteristics of vortex-induced vibration of a twin-box girder and damping effects. *J. Fluid Struct.* 103, 103282.
- Zhang, T.Y., Sun, Y.G., Li, M.S., Yang, X.W., 2020. Experimental and numerical studies on the vortex-induced vibration of two-box edge girder for cable-stayed bridges. *J. Wind Eng. Ind. Aerod.* 206, 104336.